# MATH 345 Skeleton Notes
# Unit 3: Beyond linear maps: local linearization
# Section 3.3: One-parameter maps and changes



## Vector-valued functions of one variable

### Basic definitions

A map $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ takes one scalar parameter and outputs a vector. It traces a curve.
Anatomy of a linear map lines are the special case $\mathbf{r}(t) = \mathbf{x}_0 + t\mathbf{d}$. Indeed, we already encountered parameterizations of curves when discussing parametric equations for straight lines, as in Parametric equation for a line.

**Activity**

The line $L_1$ passes through point $(3, -2, 5)$ and is parallel to the vector $\mathbf{v} = (2, -3, 4)$.
Find a parameterization of the line $L_1$.

The line passes through $(3, -2, 5)$ and has direction vector $(2, -3, 4)$, and so it parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 + 2t \\ -2 - 3t \\ 5 + 4t \end{bmatrix}.$$

**Activity**

The line $L_1$ passes through point $(3, -2, 5)$ and is parallel to the vector $\mathbf{v} = (2, -3, 4)$.
Another line $L_2$ passes through the point $(1, 3, -1)$ and is perpendicular to $L_1$. If $L_2$ also lies in the plane $z = -1$, find a parameterization of $L_2$.

Since $L_2$ lies in the plane $z = -1$, the $z$-coordinates of all the points of $L_2$ are the same, so any direction vector for $L_2$ must be of the form $\mathbf{d} = (a, b, 0)$ for some scalars $a$ and $b$. Since $L_2$ is perpendicular to $L_1$, direction vectors for $L_2$ must be perpendicular to direction vectors of $L_1$, and since $L_1$ has $(2, -3, 4)$ as a direction vector, we find that

$$\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = 2a - 3b = 0.$$

Thus $a = (3/2)b$.

Any vector solving this equation gives a direction vector of $L_2$, and so (to obtain integer values of $a$ and $b$) we can pick $b = 2$, and thus $a = 3$. So $\mathbf{d} = (3, 2, 0)$ is a direction vector of $L_2$, and since $L_2$ passes through $(1, 3, -1)$, we obtain a parameterization of $L_2$ of the form

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + 3t \\ 3 + 2t \\ -1 \end{bmatrix}.$$

The parameterizations of straight lines given above are given by vector-valued functions whose components are linear functions. But we can also parameterize curves using nonlinear functions.

**Example** As we vary $t$ over its allowed values, the vector-valued function $\mathbf{r}(t)$ traces a curve in $\mathbb{R}^n$. We will not graph functions by hand in this course. For example, consider

$$\mathbf{r}(t) = 4\cos t\mathbf{i} + 4\sin t\mathbf{j} + t\mathbf{k}, \quad 0 \le t \le 4\pi.$$

Then the curve it traces is illustrated below.

**Activity**

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(0)$.

We substitute 0 for $t$ in the equation above, obtaining that

$$\begin{aligned} \mathbf{r}(0) &= ((0)^2 - 3(0))\mathbf{i} + (4(0) + 1)\mathbf{j} \\ &= 0\mathbf{i} + 1\mathbf{j} \\ &= \mathbf{j}. \end{aligned}$$

We can also rewrite the equation $\mathbf{r}(0) = \mathbf{j}$ as $\mathbf{r}(0) = (0, 1)$.

**Activity**

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(1)$.

We calculate that

$$\begin{aligned} \mathbf{r}(1) &= ((1)^2 - 3(1))\mathbf{i} + (4(1) + 1)\mathbf{j} \\ &= (1 - 3)\mathbf{i} + (4 + 1)\mathbf{j} \\ &= -2\mathbf{i} + 5\mathbf{j}. \end{aligned}$$

Alternatively, we can write $\mathbf{r}(1) = (-2, 5)$.

**Activity**

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(-4)$.

We calculate that

$$\begin{aligned} \mathbf{r}(-4) &= (16 + 12)\mathbf{i} + (-16 + 1)\mathbf{j} \\ &= 28\mathbf{i} - 15\mathbf{j}. \end{aligned}$$

So $\mathbf{r}(-4) = (28, -15)$.

We use parameterizations mainly to describe position, velocity, and tangent direction along a curve.

## Limits and continuity

**Definition: Limit of a Vector-Valued Function** A vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ approaches the $\boxed{\text{limit}}$ $\mathbf{v} \in \mathbb{R}^n$ as $t$ approaches $a$, written

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{v},$$

provided

$$\lim_{t\to a} \|\mathbf{r}(t) - \mathbf{v}\| = 0.$$

This is the limit of a scalar-valued function, and can thus be evaluated using the methods of single-variable calculus.

In practice, calculating and simplifying the expression $\|\mathbf{r}(t) - \mathbf{v}\|$ in order to evaluate the limit

$$\lim_{t\to a} \|\mathbf{r}(t) - \mathbf{v}\| = 0$$

can be quite cumbersome. Instead, the following theorem allows us to evaluate limits much more easily.

**Theorem: Limit of a Vector-Valued Function** Let $\mathbf{r}(t) : \mathbb{R} \to \mathbb{R}^n$ be a function with component functions $r_1, \ldots, r_n : \mathbb{R} \to \mathbb{R}$. Then if

$$\lim_{t\to a} r_1(t) = v_1 \qquad \cdots \qquad \lim_{t\to a} r_n(t) = v_n,$$

and we define the vector $\mathbf{v} = (v_1, \ldots, v_n)$, then

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{v}.$$

Equivalently,

$$\lim_{t\to a} \mathbf{r}(t) = \begin{bmatrix} \lim_{t\to a} r_1(t) \\ \vdots \\ \lim_{t\to a} r_n(t) \end{bmatrix}.$$

**Activity**

Calculate $\lim_{t\to 3} \mathbf{r}(t)$ for the given vector-valued functions $\mathbf{r}(t)$.
The function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ defined by

$$\mathbf{r}(t) = (t^2 - 3t + 4)\mathbf{i} + (4t + 3)\mathbf{j}.$$

The component functions of the function $\mathbf{r}$ are $r_1(t) = t^2 - 3t + 4$ and $r_2(t) = 4t + 3$. We calculate that

$$\lim_{t\to 3} r_1(t) = \lim_{t\to 3} t^2 - 3t + 4 = (3)^2 - 3(3) + 4 = 4$$

and

$$\lim_{t\to 3} r_2(t) = \lim_{t\to 3} 4t + 3 = 4(3) + 3 = 15.$$

Thus by Limit of a Vector-Valued Function,

$$\lim_{t\to 3} \mathbf{r}(t) = \begin{bmatrix} 4 \\ 15 \end{bmatrix}.$$

**Activity**

Calculate $\lim_{t\to 3} \mathbf{r}(t)$ for the given vector-valued functions $\mathbf{r}(t)$.
The function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by

$$\mathbf{r}(t) = \left(\frac{2t-4}{t+1}\right)\mathbf{i} + \left(\frac{t^2+1}{2}\right)\mathbf{j} + (4t-3)\mathbf{k}.$$

We calculate that

$$\begin{aligned}
\lim_{t\to 3} \mathbf{r}(t) &= \lim_{t\to 3}\left[\frac{2t-4}{t+1}\mathbf{i} + \frac{t^2+1}{2}\mathbf{j} + (4t-3)\mathbf{k}\right] \\
&= \left[\lim_{t\to 3}\frac{2t-4}{t+1}\right]\mathbf{i} + \left[\lim_{t\to 3}\frac{t^2+1}{2}\right]\mathbf{j} + \left[\lim_{t\to 3}(4t-3)\right]\mathbf{k} \\
&= \left[\frac{2(3)-4}{3+1}\right]\mathbf{i} + \left[\frac{3^2+1}{2}\right]\mathbf{j} + [(4\cdot 3-3)]\mathbf{k} \\
&= \left[\frac{6-4}{4}\right]\mathbf{i} + \left[\frac{9+1}{2}\right]\mathbf{j} + [12-3]\mathbf{k} \\
&= \frac{1}{2}\mathbf{i} + \frac{10}{2}\mathbf{j} + 9\mathbf{k}
\end{aligned}$$

**Definition: Continuity of Vector-Valued Functions** A vector-valued function $\mathbf{r}(t)$ is $\boxed{\text{continuous}}$ at point $t = a$ if

$$\lim_{t \to a} \mathbf{r}(t) = \mathbf{r}(a)$$

That is, a vector-valued function is continuous at point $t = a$ when the limit of the function as $t$ approaches $a$ equals the value of the function at $a$.

**Theorem** A vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ with component functions $r_1, \ldots, r_n$ is continuous at $t = a$ if and only if each of the component functions $r_1, \ldots, r_n$ is continuous at $t = a$.

## Derivatives

Now we define the derivative of a vector-valued function from $\mathbb{R}$ to $\mathbb{R}^n$.

**Definition: Derivative of a Vector-Valued Function** The derivative of a function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ is

$$\frac{d\mathbf{r}}{dt}(t) = \mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t},$$

provided the limit exists. If $\mathbf{r}'(t)$ exists, then we say $\mathbf{r}$ is differentiable at $t$. If $\mathbf{r}'(t)$ exists for all $t$ in an open interval $(a, b)$, then we say $\mathbf{r}$ is differentiable on $(a, b)$. The vector $\mathbf{r}'(t)$ is called the tangent vector of the curve described by the function $\mathbf{r}$ at the point $\mathbf{r}(t)$.

**Remark** The derivative $\mathbf{r}' : \mathbb{R} \to \mathbb{R}^n$ of a differentiable function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ is *also* a vector-valued function.

As in single variable calculus, we will often have much more efficient methods of calculating the derivative of a function $\mathbf{r}$. But you should be able to work directly from the definition in simple examples.

**Activity**

Use the definition to calculate the derivative of the function

$$\mathbf{r}(t) = (2t^2 + 3)\mathbf{i} + (5t - 6)\mathbf{j}.$$

The derivative of the function $\mathbf{r}(t)$ is given by the expression

$$\lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}.$$

As before, we use Limit of a Vector-Valued Function to evaluate the limit. The two components of the expression

$$\frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

are

$$\frac{(2(t + \Delta t)^2 + 3) - (2t^2 + 3)}{\Delta t}$$

and

$$\frac{(5(t + \Delta t) - 6) - (5t - 6)}{\Delta t},$$

and it suffices to calculate the limits as $\Delta t \to 0$ for each component. For $\Delta t \neq 0$, expanding and canceling out like factors gives that

$$\frac{(2(t+\Delta t)^2+3)-(2t^2+3)}{\Delta t} = 4t + 2\Delta t,$$

and so

$$\lim_{\Delta t \to 0} \frac{(2(t+\Delta t)^2+3)-(2t^2+3)}{\Delta t} = \lim_{\Delta t \to 0} 4t + 2\Delta t = 4t.$$

Similarly, for $\Delta t \neq 0$ we can write

$$\frac{(5(t+\Delta t)-6)-(5t-6)}{\Delta t} = 5,$$

and so

$$\lim_{\Delta t \to 0} \frac{(5(t+\Delta t)-6)-(5t-6)}{\Delta t} = \lim_{\Delta t \to 0} 5 = 5.$$

We thus conclude that

$$\mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t)-\mathbf{r}(t)}{\Delta t} = \begin{bmatrix} 4t \\ 5 \end{bmatrix}.$$

Since Limit of a Vector-Valued Function tells us taking limits of functions from $\mathbb{R}$ to $\mathbb{R}^n$ is equivalent to taking the limits of each component function, taking the derivative of such a function is equivalent to taking the derivative of each component function.

**Theorem** If $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ has components $r_1, \ldots, r_n : \mathbb{R} \to \mathbb{R}$, then $\mathbf{r}$ is differentiable at $t$ if and only if each component function $r_1, \ldots, r_n$ is differentiable at $t$, and

$$\mathbf{r}'(t) = \begin{bmatrix} r_1'(t) \\ \vdots \\ r_n'(t) \end{bmatrix}.$$

Because of the theorem, we can apply all the techniques from single-variable calculus to differentiate functions from $\mathbb{R}$ to $\mathbb{R}^n$.

**Activity**

Find $\mathbf{r}'(t)$, where

$$\mathbf{r}(t) = e^{2t}\ \mathbf{i} + \ln(1+t)\ \mathbf{j} - (\cos t)\ \mathbf{k}$$

Using the theorem, we calculate that

$$\begin{aligned} \mathbf{r}'(t) &= (e^{2t})'\ \mathbf{i} + (\ln(1+t))'\ \mathbf{j} - (\cos t)'\ \mathbf{k} \\ &= 2e^{2t}\ \mathbf{i} + \frac{1}{1+t}\ \mathbf{j} + \sin t\ \mathbf{k}. \end{aligned}$$

If $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$, then $\mathbf{r}' : \mathbb{R} \to \mathbb{R}^n$ is also a vector-valued function of the same type, so we can consider the second derivatives $\mathbf{r}''$, i.e., the derivative of the function $\mathbf{r}'$.

**Activity**

Find $\mathbf{r}''(t)$, where $\mathbf{r}$ is as in the activity.

In the activity we calculated that

$$\mathbf{r}'(t) = 2e^{2t}\ \mathbf{i} + \left(\frac{1}{1+t}\right)\ \mathbf{j} + \sin t\ \mathbf{k}.$$

Applying the theorem, we find that

$$\begin{aligned}\mathbf{r}''(t) &= (2e^{2t})'\ \mathbf{i} + \left(\frac{1}{1+t}\right)'\ \mathbf{j} + (\sin t)'\ \mathbf{k} \\ &= 4e^{2t}\ \mathbf{i} - \frac{1}{(1+t)^2}\,\mathbf{j} + \cos t\ \mathbf{k}.\end{aligned}$$

## Differentiation rules

**Theorem** Suppose $\mathbf{u}, \mathbf{v} : \mathbb{R} \to \mathbb{R}^n$ are differentiable, $c$ is a scalar, and $a : \mathbb{R} \to \mathbb{R}$ is a scalar-valued function. Then

Sum Rule

$(\mathbf{u} + \mathbf{v})'(t) = \mathbf{u}'(t) + \mathbf{v}'(t)$

Product Rule

$(c\mathbf{u})'(t) = c\mathbf{u}'(t)$

Product Rule

$(a\mathbf{u})'(t) = a'(t)\mathbf{u}(t) + a(t)\mathbf{u}'(t)$

Dot Product rule

$(\mathbf{u} \cdot \mathbf{v})'(t) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

**Proof** We prove the dot product rule, in the two dimensional case. Let

$$\mathbf{u}(t) = u_1(t)\mathbf{i} + u_2(t)\mathbf{j}$$

and let

$$\mathbf{v}(t) = v_1(t)\mathbf{i} + v_2(t)\mathbf{j}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

Applying the sum and product rules for functions from $\mathbb{R}$ to $\mathbb{R}$, we find that

$$\begin{aligned}
(\mathbf{u} \cdot \mathbf{v})' &= (u_1 v_1)' + (u_2 v_2)' \\
&= (u_1' v_1 + u_1 v_1') + (u_2' v_2 + u_2 v_2') \\
&= (u_1' v_1 + u_2' v_2) + (u_1 v_1' + u_2 v_2') \\
&= \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'.
\end{aligned}$$

**Activity**

Find $(\mathbf{u} \cdot \mathbf{v})'$, where

$$\mathbf{u}(t) = \cos t \ \mathbf{i} + \sin t \ \mathbf{j} + t \ \mathbf{k}$$

and

$$\mathbf{v}(t) = t \ \mathbf{i} + \ln t \ \mathbf{j} + \mathbf{k}.$$

We calculate that

$$\begin{aligned}\mathbf{u}^{'} &= (\cos t)^{'}\ \mathbf{i} + (\sin t)^{'}\ \mathbf{j} + (t)^{'}\ \mathbf{k}\\ &= -\sin t\ \mathbf{i} + \cos t\ \mathbf{j} + \mathbf{k}\end{aligned}$$

and

$$\begin{aligned}\mathbf{v}^{'} &= (t)^{'}\ \mathbf{i} + (\ln t)^{'}\ \mathbf{j} + (1)^{'}\ \mathbf{k}\\ &= \mathbf{i} + (1/t)\ \mathbf{j} + 0\ \mathbf{k}.\end{aligned}$$

Thus

$$\mathbf{u}\cdot\mathbf{v}^{'} = \begin{bmatrix}\cos t\\ \sin t\\ t\end{bmatrix}\cdot\begin{bmatrix}1\\ 1/t\\ 0\end{bmatrix} = \cos t + \frac{\sin t}{t}$$

and

$$\mathbf{u}^{'}\cdot\mathbf{v} = \begin{bmatrix}-\sin t\\ \cos t\\ 1\end{bmatrix}\cdot\begin{bmatrix}t\\ \ln t\\ 1\end{bmatrix} = -t\sin t + \ln t\cos t + 1.$$

So

$$
\begin{aligned}
(\mathbf{u}\cdot\mathbf{v})' &= \mathbf{u}\mathbf{v}' + \mathbf{u}'\mathbf{v} \\
&= \left(\cos t + \frac{\sin t}{t}\right) + (-t\sin t + \ln t\cos t + 1). \\
&= \cos t(1+\ln t) + \sin t\,(1/t - t) + 1.
\end{aligned}
$$

**Remark** Note that in the activity, since $\mathbf{v}$ is only well-defined for $t > 0$, the function $\mathbf{u}\cdot\mathbf{v}$ is also only well-defined for $t > 0$, as is its derivative.